

SunshineCTF: Ghost in the Thread 1

Overview

This challenge started with a Virtual Disk Image (VDI) file which you had to run in VirtualBox. When you opened the file, you were expected to find the evidence of compromise. By searching through logs, and following paths, a suspicious program was found which gave the flag.

Methodology

In the Linux environment I was given, I immediately navigated to the logs section in the path `/var/log/`. It was there that I found a custom folder, `sunchan`. When I checked the options there were two log files to dig through, `thumbnail.log` and `uploads.log`.

```
sunchan:/var/log/sunchan$ ls
-rw-r--r-- 8.0k root 26 Sep 06:51 thumbnail.log
-rw-r--r-- 17k root 26 Sep 06:51 uploads.log
```

I initially used `grep` to find more information, such as “password,” “error,” and flag indicators like “sun{.” This did not turn up with anything. I finally thought to turn back to the challenge name and searched “ghost” in both of the logs, which resulted in one response:

```
sunchan:/var/log/sunchan$ grep -Ei "ghost" thumbnail.log
2026-08-14T18:42:12Z event=render.start job_id=4182 upload_id=u-7f31 renderer=ghostscript executable=/usr/bin/gs pid=2371 uid=1002 input=/srv/sunchan/uploads/po/184726.pdf
sunchan:/var/log/sunchan$
```

This was interesting, there was an executable called `ghostscript`, and it linked a path to a location `/srv/sunchan/uploads/po/184726.pdf`.

```
sunchan:/srv/sunchan/uploads/po$ cat 184726.pdf
%!PS-Adobe-3.0
%%Title: Folding guide
%%Creator: SunChan upload fixture
%%Pages: 1
%%BoundingBox: 0 0 612 792
%%Document-ID: po-184726
%%Incident-Replay: exploit trigger removed; stage metadata preserved
%%Stage2-Action: download
%%Stage2-URL-Base64: aHR0cDovL2Fzc2V0cy5zdW5jaGFuLmludmFsaWQucmVzb3UyY2VzL2dzLXJlc291cmNlLmJpbG==
%%Stage2-Destination: /var/tmp/sunchan-downloads/gs-resource.bin
%%Stage2-SHA256: 3cf1f62fe43202b9d0fc1ded2776fe7d40001d3fd6c8d23b53550d8edf0f3500
%%EndComments
/Helvetica-Bold findfont 20 scalefont setfont
72 720 moveto (Folding Guide) show
/Helvetica findfont 11 scalefont setfont
72 690 moveto (1. Fold the square diagonally.) show
72 670 moveto (2. Open it and fold the opposite diagonal.) show
72 650 moveto (3. Bring all four corners toward the center.) show
newpath 72 520 moveto 180 628 lineto 288 520 lineto 180 412 lineto closepath stroke
showpage
%%EOF
```

I used cat to open the file and view what was inside. The most interesting thing here was this Stage2-Action and Stage2-Destination section. Since there was another location linked, I went to investigate it. Note the Document-ID in this image.

```
sunchan:/var/log/sunchan$ /var/tmp/sunchan-downloads/gs-resource.bin
SunHash proof-of-work benchmark simulator (offline)
Document-ID: po-184726
sunftfw_hacked_by_offboarders}
sunchan:/var/log/sunchan$ _
```

I went to the file and tried running it, and first asked for a Document-ID. I inputted the Document-ID we got earlier, and finally the flag popped up.